PS No.	9
Project Title	**College Attendance Analyzer**
Problem Statement	Analyse student attendance from a CSV file to calculate attendance percentage, find students below the 75% requirement and show attendance patterns. **Language:** Python 3 **Concepts:** Pandas, CSV, Matplotlib, Filtering **Note:** CSV columns: Name, Subject, Classes_Held, Classes_Attended. Minimum required attendance is 75%.
Test Cases / Case Statements (4 cases)	See table below
Bare Minimum Requirements	1. Write the solution in Python 3 (one .py file) and run it from the console. 2. Load the attendance CSV file using Pandas. 3. Calculate the attendance percentage of every student. 4. Filter and display students below 75% using boolean filtering. 5. Plot at least two Matplotlib charts (bar and pie).
Instruction	1. Understand the problem statement and define the problem clearly. 2. Develop the solution using proper logic and programming concepts. 3. Implement all required features and pass the given test cases. 4. Test and debug the project to ensure correct output. 5. Prepare the documentation/report and upload the code and report to GitHub. 6. Complete and submit the project within the given deadline.

INPUT	EXPECTED OUTPUT
Test Case 1: Normal Input	
attendance.csv: Aman,Python,40,36 Riya,Python,40,28 Karan,Python,40,30 Menu option 1 (Percentage)	Aman: 90.00% | Riya: 70.00% | Karan: 75.00%
Test Case 2: Boundary / Minimum Input	
attendance.csv: Karan,Python,40,30 Menu option 2 (Shortage list)	No student below 75%. Karan (75.00%) is eligible.
Test Case 3: Invalid Input	
attendance.csv: Aman,Python,40,45	Row 1 invalid: attended (45) cannot exceed held (40). Row skipped.
Test Case 4: Different Valid Scenario	
Test Case 1 rows plus: Aman,Maths,50,45 Menu option 3 (Subject-wise average)	Python: 78.33% | Maths: 90.00% Bar chart displayed